Explaining Without Predicting: Statistical Anomaly Detection and Precedent Retrieval for Verifiable Financial Alerts

Henrique J. S. Santos[1], Luís Gomes[1], and Rafael Silva[1]

Instituto Superior de Engenharia do Porto (ISEP), Porto, Portugal
1180934@isep.ipp.pt

Abstract. Retail investors face a continuous stream of price movements and financial news without the interpretive tools available to institutions. We present InvestiGator, an explainable financial alerting system for the US equity market, built exclusively on free data sources and deployed in continuous operation. The system answers three questions a retail investor faces when a price moves, and attaches to every alert the evidence that supports it: whether the movement is unusual for that company, whether its origin lies with the company or the market, and whether comparable precedents exist and what followed them. A design constraint runs through the architecture: the system issues no price forecasts, because a statement about a past event can be checked against the record at the moment it is read, whereas a forecast can only be assessed after the decision it informs has been taken. Each component is evaluated against transparent baselines declared in advance. Volatility-normalised detection fires at a near-constant rate across companies of very different volatility, with a spread of 0.015 against 0.344 for a fixed-threshold rule, and outperforms two learned detectors under an equivalent causal protocol. Semantic retrieval beats the sector base rate in all five sectors evaluated and retains its margin under the temporal constraint the deployed system faces. A supervised triage model *does not* outperform a volatility-only baseline: added to a per-company lookup table, the headline text contributes +0.012 PR-AUC, an upper bound rather than a usable quantity. We report that negative result with the same prominence as the positive ones, and locate it: the information the text carries separates companies and periods, where the product needed it to separate news items.

Keywords: Explainable AI · Financial alerting · Anomaly detection · Semantic retrieval · Retail investors

1 Introduction

Buying shares has become a low-friction operation. Most American adults now hold equities, directly or through funds and retirement plans [1], in a market whose capitalisation exceeds 62 trillion US dollars [2]. Ease of access has not extended to interpretation. When a price moves, the retail investor has limited means of determining what that movement means.

Three concrete questions remain unanswered at the moment an investor observes a price move. First, *is the movement unusual?* A four per cent change is a rare event for a low-volatility company and an ordinary day for a volatile one. Second, *does the movement originate with the company or with the market?* A decline that tracks the index and an isolated decline are different situations. Third, *are there precedents, and what followed them?*

Free applications display the percentage change; the few that go further deliver a generated summary without the evidence behind it. Professional terminals answer all three, at subscription costs incompatible with this audience. The gap is not one of data availability, which is ample and free, but of verifiable context built on top of it.

This paper presents InvestiGator, a deployed system that answers the three questions and exposes the evidence for each answer, and reports the evaluation of its components. As an Artificial Intelligence Engineering contribution, the work lies not in formulating a new algorithm but in the selection, integration and critical evaluation of established techniques until they constitute a functioning, explainable and reproducible system, including the result that contradicts the initial hypothesis.

A design constraint. The system produces *no price forecasts.* It gives no directional indication, issues no buy or sell recommendations and displays no price targets. This has two justifications. The first is verifiability: a statement about a past event can be confirmed against the record at the moment it is read, whereas a forecast is only assessable after the investor's decision has been taken. The second is theoretical: the semi-strong form of the efficient market hypothesis [3] makes systematic anticipation from public information unlikely. This work neither tests that hypothesis nor relies on it to establish any conclusion.

2 Related Work

Each component the system uses exists in the literature in isolation. What does not exist is their combination under a zero-cost constraint, with the evidence delivered to the user at the moment the claim is made.

Anomaly detection in financial series. Surveys of anomaly detection [4] distinguish statistical from learned approaches. Isolation Forest [5] and the Local Outlier Factor [6] define rarity geometrically and were designed for spaces with several interacting variables. Financial return series exhibit volatility clustering, the property that the ARCH and GARCH families formalise [7,8]: periods of agitation follow periods of calm. That structure is temporal rather than geometric, which conditions the comparison reported in Sect. 4.

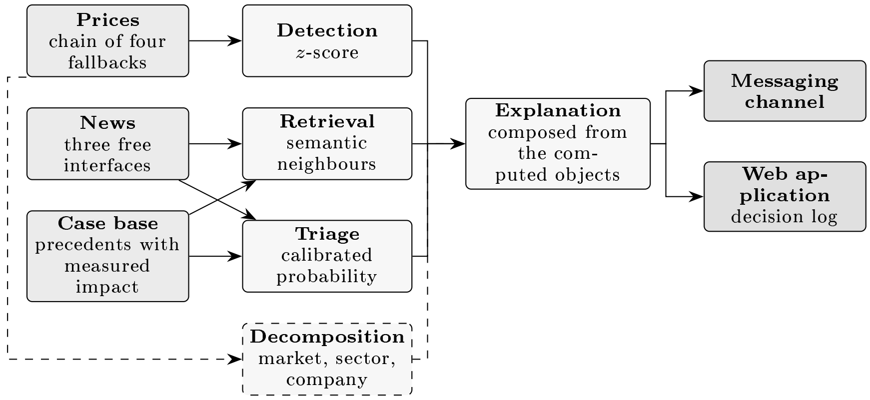

Fig. 1. System components and their connections. Decomposition is dashed because it is the only component that does not take part in every evaluation: without a price movement there is nothing to apportion.

Sentence representation and retrieval. Contextual encoders [9] transformed text representation, but their raw sentence embeddings are poorly suited to similarity search. Sentence-BERT [10] adds the *training objective* that places sentences in a space where distance means similarity, which is the property retrieval requires. Domain-specific financial encoders exist [11,12]; the natural assumption that they transfer to this task is tested in Sect. 4 and does not hold.

Event studies and case-based reasoning. Measuring what followed an event is the subject of the event-study method [13], and sensitivity estimation benefits from shrinkage towards a reference [14]. Presenting past cases rather than a rule is the premise of case-based reasoning [15]. The system combines the two: it retrieves semantically close cases and reports the price behaviour *measured* after each, rather than a prediction derived from them.

Explainability and trust. The literature distinguishes explaining an opaque model *post hoc* [16,17] from building a system whose reasoning is inspectable by construction [18]. This work takes the second route, and the reason is one of risk: trust in automated systems fails in both directions [19], and the mere presence of an explanation increases acceptance of incorrect answers [20]. A generated sentence is an assertion; a sentence composed from computed objects is an assertion with the evidence attached, in which every displayed value points back to the measurement that produced it.

3 The InvestiGator System

Figure 1 shows the components and their connections. Sources on the left, methods in the centre, delivery channels on the right.

3.1 The Composition Rule

One construction rule runs through the whole implementation: the text delivered to the user is *composed* from the objects the components produced, and is not written by a language model. Each displayed value comes from the object that computed it, so the alert cannot state a value other than the one the system obtained. Fidelity is therefore a property of the construction, not the outcome of a subsequent check.

The system does implement a grounded generation layer, in which every sentence must cite an evidence record and a verifier rejects any sentence whose numbers do not belong to the records it cites. That layer is *implemented and evaluated but not exposed* in the deployed product; it is reported separately in Sect. 4 for that reason.

3.2 Detection: What Counts as Unusual

A movement is unusual relative to the company's own recent behaviour, not to a fixed percentage. For each trading day t, the log return $r_t = \ln(P_t/P_{t-1})$ is compared with the mean and standard deviation of the preceding window:

$$z_t = \frac{r_t - \mu_{t-1}}{\sigma_{t-1}}, \qquad \mu_{t-1}, \sigma_{t-1} \text{ over } [t-w,\, t-1]. \tag{1}$$

The day being assessed is *excluded* from the window that defines its own norm. Without that exclusion the day would contribute to the mean and deviation against which it is compared, attenuating its own anomaly. The same discipline forbids the use of any information later than the day assessed, and an automated test enforces it by mutating the future of an example and requiring the inputs to remain identical while the label changes.

3.3 Decomposition: Company or Market

The observed return is apportioned into a market component, a sector component orthogonal to the market, and a company-specific residual:

$$r = \beta_m\, r_m + \beta_s\, \tilde{r}_s + \varepsilon. \tag{2}$$

Sensitivities are estimated only on data *prior* to the day being explained, and are shrunk towards a reference by precision weighting [14] rather than clipped at an arbitrary bound. The distinction is not cosmetic: a clipping rule that falls back to a unit sensitivity whenever the estimate leaves a fixed range attributes the whole of an unusual movement to the company precisely on the days when the estimate is least reliable, which are the days the alert is about.

3.4 Retrieval: Precedents with Measured Outcomes

Each headline is encoded into a 384-dimensional sentence embedding [10], and precedents are retrieved by cosine similarity from a case base of 38,214 entries. What distinguishes retrieval here from a similarity search is that every retrieved case carries the price behaviour *measured* after it, over +1, +3 and +5 trading days, aligned to the first session on or after the news date. Precedents are shown individually, with date, similarity and outcome, and never summarised into an average that could be read as a forecast.

3.5 Decision Policy

Communicating every candidate is equivalent to communicating nothing. A channel delivering fifteen messages a day stops being read, and the system loses the usefulness that justifies it. Five decision points act in sequence: the headline must name the company; it must be recent; a precedent must exceed a similarity floor; the triage score must exceed a floor; and a global budget of five alerts per day must not be exhausted. Over 1–3 September 2026 the system evaluated 743 distinct headlines from twelve companies and delivered 15 alerts to eight of them.

That total of fifteen is the ceiling the budget imposes rather than a measurement: the budget was exhausted on all three days. What the per-company distribution measures is the ordering *within* that ceiling, which the policy does not fix, since no company holds a quota of its own.

4 Evaluation

Every comparison is against a baseline chosen and declared *before* the measurement, which is the condition under which a result can come out negative. Each study states the question, the procedure and the alternative, the result obtained, and what the result does *not* allow one to conclude.

4.1 Detection Is Consistent Across Companies

The primary argument is label-free: a detector that measures rarity relative to each company's own norm should fire at a comparable rate across companies of very different volatility. Over three years of daily data for fifteen large-cap US tickers, the spread between the most and the least flagged company is 0.015 for the rolling z-score against 0.344 for a fixed three per cent rule. Against an approximate extreme-move label the z-score reaches $F_1 = 0.516$ against 0.218 for the fixed rule.

Under an equivalent causal protocol the same method also outperforms two detectors designed for this task, with F_1 of 0.530 against 0.269 for the Isolation Forest [5] and 0.280 for the Local Outlier Factor [6]. The literature situates that outcome: both learned detectors define rarity geometrically and were designed for multivariate spaces, whereas the problem here presents a single return series per company in which the relevant structure is temporal. When the assumption embedded in a simple method coincides with the actual structure of the problem, the margin for improving on it is small.

What this does not establish. The deployed configuration is not the best measured within its own family: an exponentially weighted estimator reaches $F_1 = 0.664$ and a sixty-day window 0.678, against 0.516 in production. The window was kept for responsiveness and the equally weighted deviation for explainability. Those are decisions, not results, and the paper reports them as such.

4.2 Retrieval Beats the Base Rate in Every Sector

Relevance is approximated by sector membership, a verifiable proxy rather than the property one would wish to measure, and the query's own company is excluded so that the task cannot be solved by ticker matching. Over 3,714 real headlines across five sectors, MiniLM reaches a cross-ticker precision@5 of 0.514 and the larger MPNet 0.538, both above the lexical baseline of 0.346; recency, at 0.126, is worse than random.

The aggregate is not, however, the right way to state the result, and we state the weaker claim the data supports. Almost half the corpus is technology news, which makes available a strategy that uses no model and does not even read the query: always return technology headlines. Its precision equals the fraction of technology queries, 0.467, above the lexical baseline. That strategy is useless as a product, since it is right on every query in one sector and wrong in the four others, but its existence shows that the aggregate partly measures corpus composition. The comparison that survives the objection is made *within* each sector, where the method exceeds the corresponding reference value in all five (Fig. 2).

Two checks bound the claim, and the right-hand panel of Fig. 2 shows them. On the multi-year corpus of approximately eighty thousand headlines, precision@5 rises to 0.595 while the base rate rises in the same proportion: the method neither degrades nor improves with scale. Imposing the constraint the deployed system actually faces, that a precedent must predate the query, lowers precision to 0.513 and the base rate to 0.259, moving the margin by 0.008. The method keeps its advantage when forced to operate as it does in production.

Theme is not direction. Agreement between the direction of the query's movement and that of its precedents is 0.708 against a chance value of 0.688. Two headlines can concern the same subject and the prices move in opposite directions. The consequence for the product is the decision to present precedents individually, with date and outcome, rather than reducing them to an average that could be read as a forecast.

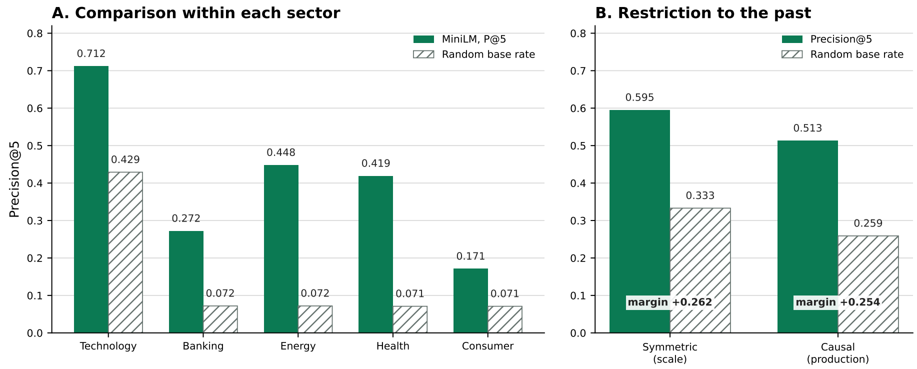

Fig. 2. Left, per-sector cross-ticker precision@5 against the sector base rate: the absolute margin is largest in the most abundant sector, while in the scarce sectors, where the reference value is near 0.072, the method reaches several times that value with a smaller absolute margin. Right, the same measurement under the constraint the deployed system faces, that a precedent must predate the query: precision falls to 0.513 and the base rate to 0.259, so that the margin over chance moves by 0.008.

Table 1. Triage, PR-AUC on the held-out test block (prevalence 0.378). The baseline declared in advance was the volatility-only model.

Model	PR-AUC	Brier
Volatility only (baseline)	**0.542**	0.218
Company and day context	0.538	0.224
Context and headline text	0.496	0.229
Gradient boosting	0.469	0.228
Headline text only	0.439	0.240
Always alert (floor)	0.378	0.622

4.3 The Negative Result: Text Does Not Beat Volatility

The triage task is to decide, before delivery, which headlines justify interrupting the reader. We built a dataset of 79,753 examples from the public FNSPID corpus [21] and price series, with a single-day temporal split and an embargo, and trained six families. Table 1 reports them.

No variant incorporating the headline text outperforms the volatility-only baseline. The result survives three checks. The first repeats the comparison granting the text the best available conditions, with tuned regularisation, a PCA-reduced text block and a finance-domain encoder, and raises its best value to 0.533 without reaching the baseline. The second resamples by groups formed of company and day, and keeps the interval excluding zero. The third repeats the measurement for nine alternative combinations of label threshold and horizon, and volatility matches or exceeds the text model in all nine.

The domain encoder is the weakest of all. The finance-specific encoder reaches 0.420, the lowest value among the models evaluated. The artefact assessed is the public checkpoint `ProsusAI/finbert`, applied by mean-pooling its vectors, which is the way to use it without further training. It is an encoder tuned for sentiment classification and not for placing sentences in a space where distance means similarity, and it is precisely that training objective that Sentence-BERT adds [10]. The measurement establishes that this domain model, used in this way, performs worse than a general model trained for similarity; it does not establish that domain knowledge is irrelevant to the task.

4.4 Locating the Negative Result

Placing representations side by side answers which of them is better. The question with more use to whoever builds a system is different: does the text add to what is already known? Measured against a per-company lookup table, which contains everything the model knows about the company and nothing about the news item, the answer is affirmative: the headline adds +0.012 PR-AUC, with an interval between +0.004 and +0.020 that excludes zero. It is the only occasion on which a text representation shows a detectable increment in this task.

The value falls below the 0.02 threshold fixed before the measurements, and that threshold applies in both directions. What is established is an *upper bound* on the effect of the text, not a quantity the system can use. Three further measurements prevent this increment from reopening the verdict: the sum does not exceed volatility alone; the increment does not survive the metric that describes the product, since precision within the five-alert budget remains at 0.662 with and without text; and the ability to separate news items from the same company remains at chance level.

What these measurements add to the negative answer is not a caveat but a *location*: the information the text carries separates companies and periods, where the product needed it to separate news items.

An engineering consequence. A per-company lookup table of thirteen constants, containing no information from any headline, reaches 0.534 against 0.538 for the deployed variant, and the same precision within budget. The deployed model *is*, in effect, a lookup table. That finding is what led to the policy change reported next.

4.5 The System in Operation

The system has run continuously on free infrastructure, monitoring twelve companies. Over the periods documented it delivered 367 messages, between 9 July and 13 August 2026, and recorded 36,925 triage decisions over a wider window, while the precedent base accumulated 38,214 cases with measured impact.

A measured policy change. Once the triage model was found to be a lookup table, it was changed from a veto into a ranking, under a global daily budget. The effect was measured, with a control. In news alerts, which the budget governs, concentration in the three most alerted companies falls from 0.713 to 0.480, with the interval of the difference excluding zero, and the number of companies reaching an alert rises from seven to nine of twelve. Price-movement alerts cross the same period, the same companies and the same conditions, and the budget does *not* count them: in that series concentration moves from 0.449 to 0.485, with the interval containing zero. The governed quantity moved and the ungoverned one did not.

Grounded generation, measured separately. For the layer that is implemented but not exposed, two quantities are measured because they answer different questions. An adversarial corpus of 23 attacks, each of which must be rejected, measures the guard: 23/23 are blocked. A set of 8 faithful control sentences, each of which must pass, measures whether the guard is merely refusing everything: 8/8 are admitted. The second is the load-bearing figure, since a guard that rejected all text would score perfectly on the first.

5 Discussion and Limitations

No controlled user study. This is the heaviest gap. The system rests on the hypothesis that an explanation accompanied by its evidence leads a non-specialist to a better decision, and that hypothesis has not been submitted to an experimental design. The protocol exists and has not been executed. What does exist is feedback collected in a real context: readers of the public channel classify the alerts they receive, under analysis rules fixed before the first vote. That feedback is observational: participation is self-selected, no group received the price change without the accompanying explanation, and the channel is public, so it is not known whether respondents belong to the audience this work assumes. The founding hypothesis concerns a better decision, which is a different quantity, and it remains untested.

Latency is bounded by the sources, not by the cycle. Between publication and detection the median is 353 minutes, and between detection and delivery it is five seconds, with a ninetieth percentile of sixteen. The time is entirely in discovery, and shortening the polling cycle did not reduce it: free company-news interfaces are not real-time channels, and the most recent headline in a feed is not the most recent *relevant* one. The system observes, by construction, information that is no longer new. That condition aggravates rather than attenuates the expectation behind the negative result of Sect. 4.

Labels are verifiable approximations. Sector membership approximates relevance; a return threshold approximates materiality; a percentile approximates rarity. Each is verifiable and none is the property one would wish to measure. Closing that distance requires a human-annotated sample, which does not exist.

Survivorship and coverage. The companies were selected in 2026 and evaluated over the past, so they survived by construction. News coverage is incomplete and uneven: at least one headline was captured on 88.5% of unusual days, which is an upper bound, since it asks whether *a* story existed and not whether it was the right one.

6 Conclusion

InvestiGator shows that useful financial alerting for retail investors can be transparent and reproducible at once: a volatility-normalised detector that explains every flag, a retrieval engine that returns analogous precedents and reports their *measured* outcome as evidence rather than prediction, a decomposition that separates the company from the market, and a calibrated triage model that honestly fails to beat a volatility-only baseline with text.

The most transferable observation is that last one, and it concerns evaluation rather than modelling. A component evaluated in isolation, over the distribution of its training set, and deployed behind filters that change that distribution, has not been evaluated on the distribution it will see. The metric was correct and the question it asked was not the one the product needed: PR-AUC over a full test set is dominated by variation *between* companies, whereas the product required separation *within* a company on a given day. Detecting that required comparing the deployed model against a lookup table of thirteen constants, which is a comparison worth making before concluding that a learned component earns its place.

Future work is a market-adjusted impact-magnitude study over the multi-year precedents, the controlled user study whose protocol is already prepared, an annotated sample that would anchor all three labels at once, and closing the text gap with richer input than headlines.

Reproducibility. All results are produced by versioned scripts over frozen artefacts, with fixed seeds. The trained model, its calibration and the evaluation outputs are kept as versioned artefacts, and each figure in this paper is regenerated by the script that produced the corresponding measurement.

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